PRAKHAR DWIVEDI

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SUMMARY

Data Scientist / AI Engineer with a B.Tech from **NIT Andhra Pradesh**. Experience spans ML modeling, deep learning, and AI systems (**LLM**, **RAG**) — built through a Data Science internship at **IIM Indore** and self-directed projects in **computer vision**, **NLP**, and **Agentic AI**. Skilled in **Python**, **Deep Learning**, and **Generative AI**, with model deployment via **Hugging Face Spaces**.

EDUCATION

National Institute of Technology, Andhra Pradesh — CGPA: 7.91 / 10
Bachelor of Technology in Engineering — 2021 – 2025
Lions School, Lal Diggi, Mirzapur, UP — Intermediate (Science) — CGPA: 10 / 10
Lions School, Lal Diggi, Mirzapur, UP — High School (Science) — 94.8%

EXPERIENCE

Data Science Intern — IIM Indore (On-Site) — May 2023 – Jul 2023
Projects: (a) S.I.R. ML Model (b) Movie Data Viz

- Built a disease-spread **forecasting model** using the S.I.R. framework — fitting linear regression to epidemiological data and solving the resulting differential equation numerically — achieving **R^2 of 0.91** in predicting infection and recovery rates.
- Analysed genre-vs-budget relationships across ~560 movies by building a multi-plot visualization framework in R and ggplot2.
- Independently extended this model in 2026, designing a hybrid physics-ANN architecture — differential equation as physics-based prior with an ANN correction layer — achieving **R^2 of 0.89 (RMSE 19.4)** on non-linear data.

Web Developer — Counselxperts (College Startup) — Jul 2023 – Sep 2023

- Onboarded **100+ new users** during initial launch by shipping a full-stack website.
- **Reduced transaction time by ~40%** by designing responsive, cross-platform payment UI components.
- Cut new-developer setup time by building a reusable React/JavaScript starter framework adopted across the team.

Web Developer — Sawari (College Startup) — Dec 2022 – Feb 2023

- **Cut payment processing time by ~60% (5 min → 2 min)** by integrating a Google Forms-based payment workflow directly into the company website.
- Increased site engagement by designing and deploying a targeted marketing website section.

Operations & Strategy Associate — Competitive Conqueror (College Startup) — Aug 2022 – Oct 2022

- **Generated ₹10,000+ profit** in **8 weeks** with **zero capital** by launching and operating a self-funded venture, from brand identity through go-to-market.

On-Site Intern — Core Metal Kraft Limited (L&T / Schneider Electric Partner) — Jul 2022 – Aug 2022

- Reduced electromagnetic interference risk in LT panel design by redesigning compartment layout with the senior CAD team, reviewing 10+ engineering blueprints.

PROJECTS

CNN-Ensemble Model for Real-Time Surface Roughness Prediction [Link] [Video]

- Achieved **90% predictive accuracy** and an **R^2 of 0.90** by combining **CNN** feature extraction with a stacked **XGBoost/SVM ensemble**.
- Delivered **live roughness prediction** with **under 1-minute latency** by deploying the model into a real-time computer vision GUI system.

Audio-to-Meeting-Minutes Generator (LLM/NLP) [Link] [Deployed]

- **Automated ~70% of manual documentation effort** by building an LLM-based pipeline using **LangChain**, **Hugging Face**, and **OpenAI Whisper**.
- Delivered structured meeting minutes with defined decisions and tasks in **under 5 minutes** by **automating speech-to-text transcription end to end**.

Agentic AI Portfolio Lead Capture Assistant (Local RAG) [Link] [Deployed]

- **Improved token efficiency by ~50%** by building a local **RAG pipeline** with **section-based chunking** using **Ollama** and **Transformers**.
- **Automated 100% of lead-capture** by integrating **autonomous function-calling tools** with an **SMTP notification workflow**.

SKILLS

Programming: Python, R | **Generative AI:** LLM Engineering, Context Engineering, Vector Embedding, NLP, Model Evaluation
Machine Learning: TensorFlow, PyTorch, Scikit-learn, CNN, ANN, Deep Learning | **Computer Vision:** OpenCV | **Tools:** Git, GitHub
Data Analysis: Pandas, NumPy, SciPy, SQL, Matplotlib, ggplot2, Statistical Analysis | **Agentic AI:** Agents SDK, RAG, Tool Calling,
Deployment: FastAPI, Gradio, Hugging Face Spaces | **Web Development:** React.js, Node.js, Express.js

CERTIFICATIONS & ACCOMPLISHMENTS

Machine Learning Specialization (Stanford Online/DeepLearning.AI) | **Agentic AI Fundamentals, Generative AI Fundamentals** (LinkedIn Learning) **[Link]**
National Science Olympiad (NSO): 1st (Class), 242nd (Int'l) | **International Olympiad of Mathematics (IOM)**: 2nd (Class), 274th (Int'l)